

On the Effect of Prior Specification in Bayesian Symbolic Regression

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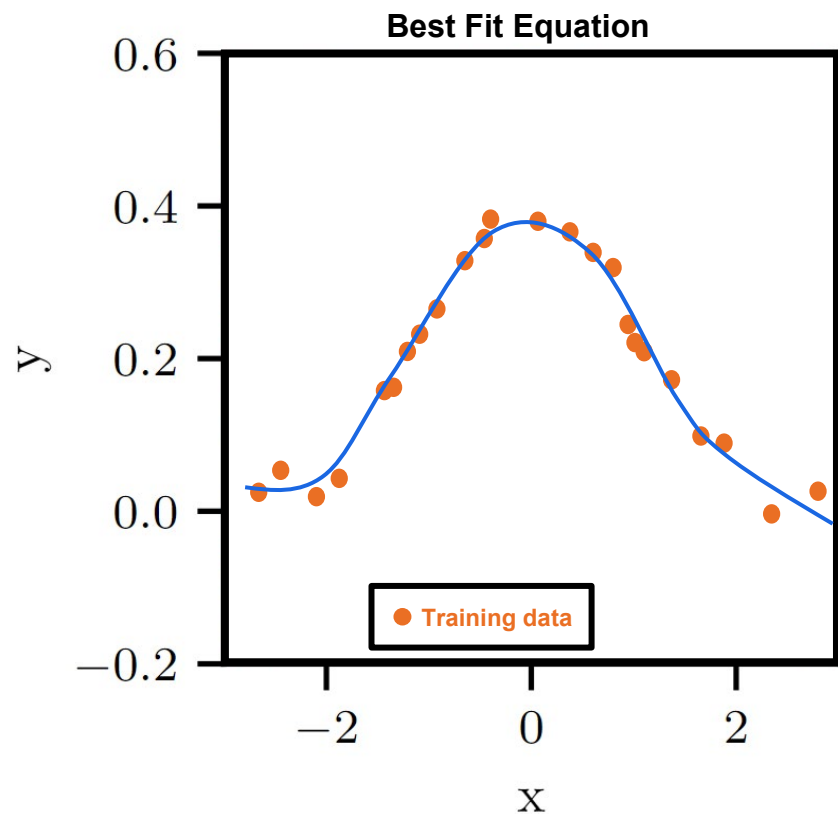
1. NASA Langley Research Center

2. Analytical Mechanics Associates

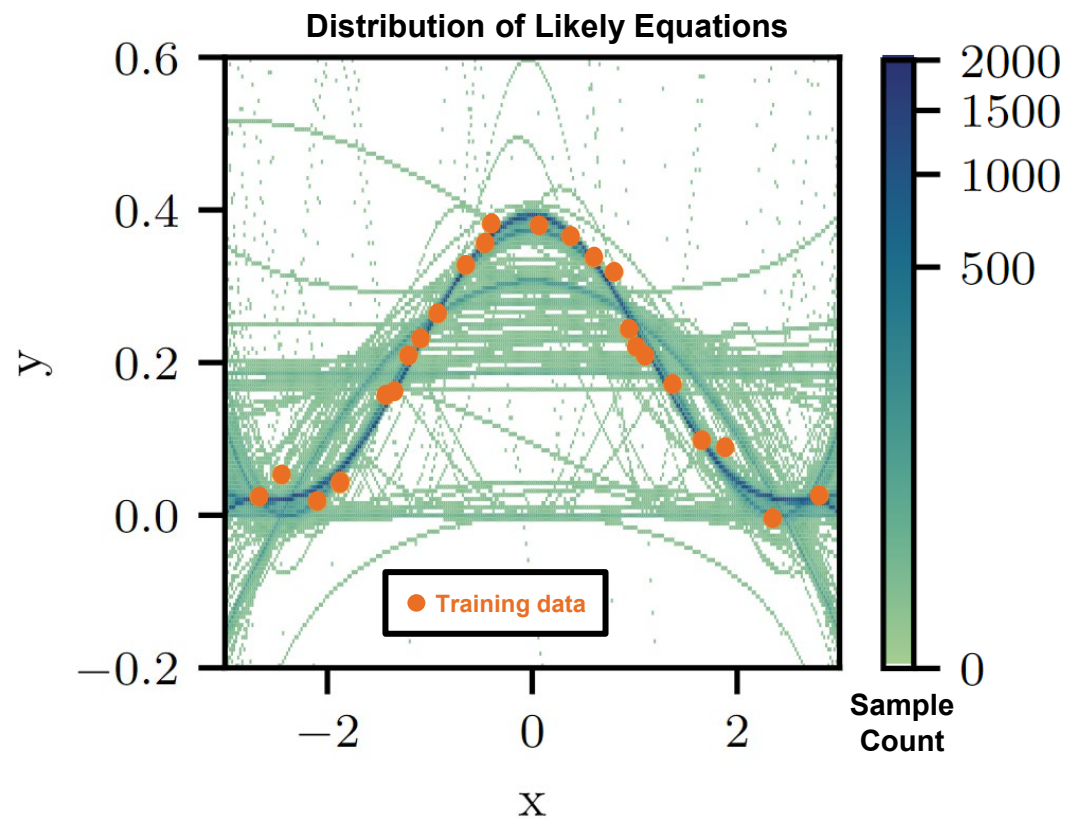
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Bayesian Symbolic Regression

Standard Symbolic Regression



Bayesian Symbolic Regression



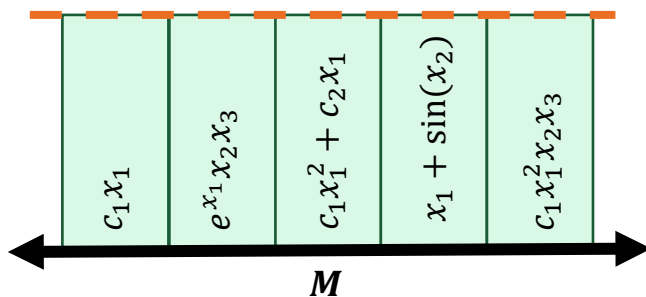
Bayesian Symbolic Regression

- Applying Bayesian Inference to Symbolic Regression

$$\overset{\text{Posterior}}{P(M|D)} \propto \overset{\text{Likelihood}}{P(D|M)} \overset{\text{Prior}}{P(M)}$$

Prior

what we know about equations
before seeing any data

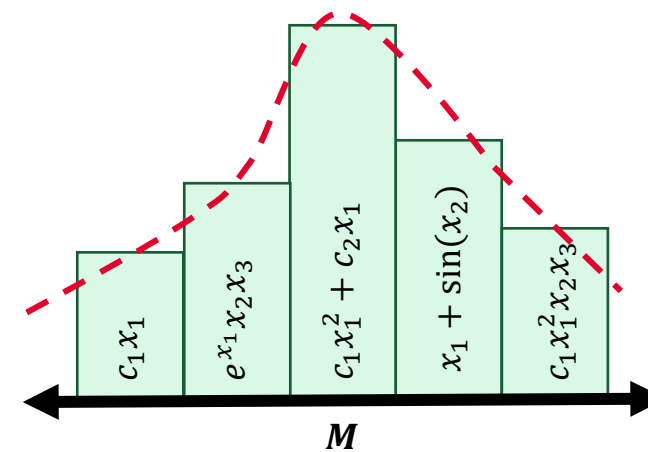


PYSIPS
Symbolic Inference via
Posterior Sampling



Posterior

Probability of model given data

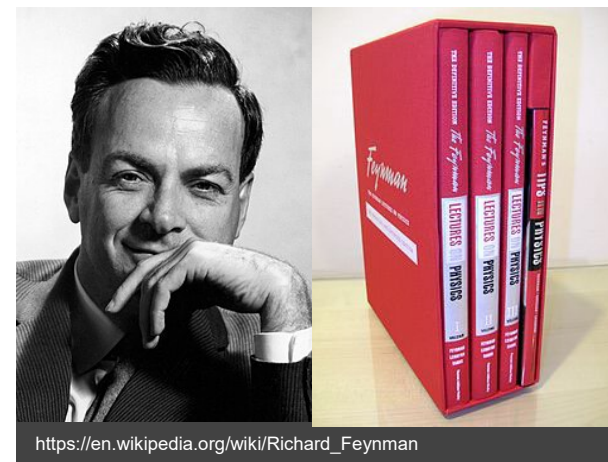


Prior Construction

1. Identify a corpus of equations that represents the subject matter of interest
2. Identify features/statistics of interest
 - Operator occurrence
 - Equation complexity
3. Build prior to encapsulate those feature(s)

Simplest: (improper) uniform prior
all equations are equally probable

Feynman Equations



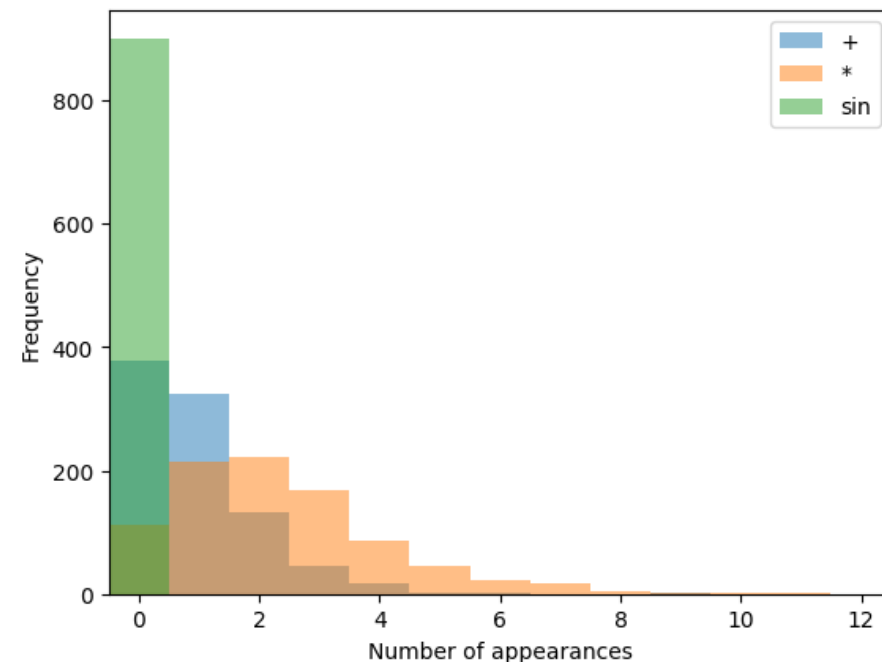
Wikipedia

List of scientific equations
named after people

https://en.wikipedia.org/wiki/List_of_scientific_equations_named_after_people

Bayesian Machine Scientist (BMS) Prior

- Random graph theory
- **Goal:** match appearance rates of each operator
- **Fitting:**
 - Matches first 2 moments of distribution
 - Inverse fitting by iteratively sampling the prior distribution
- **Result:** More common operators are more likely



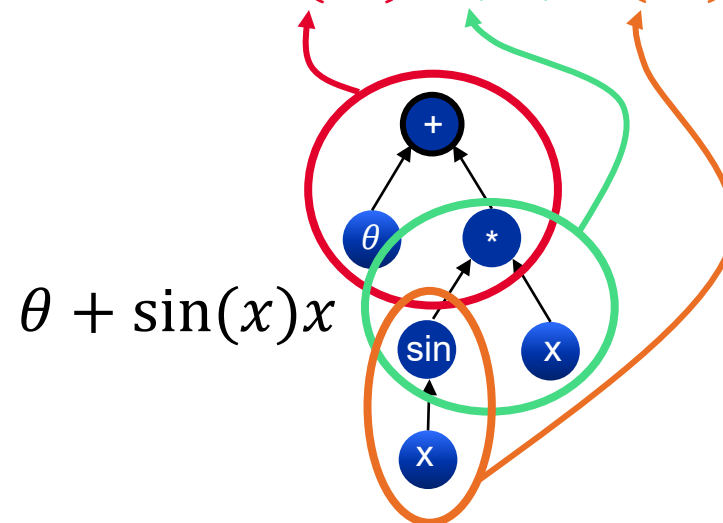
$\log P(M)$ for representative equations (higher = more likely)

	Uniform	BMS
$\sin(x_0) + \sin(x_1)$	0	-2.80
$\sin(\sin(x_0 + x_1))$	0	-2.80

Katz Backoff Prior

- N-gram language model
- **Goal:**
 - match appearance rates of operator groups
- **Fitting:**
 - Bi-gram implementation
 - Back off to unigrams when needed
 - Telescoping product of appearance rates
- **Result:**
 - More common operator groups are more likely

$$P(M) = P(\dots)P(\dots)P(\dots)$$



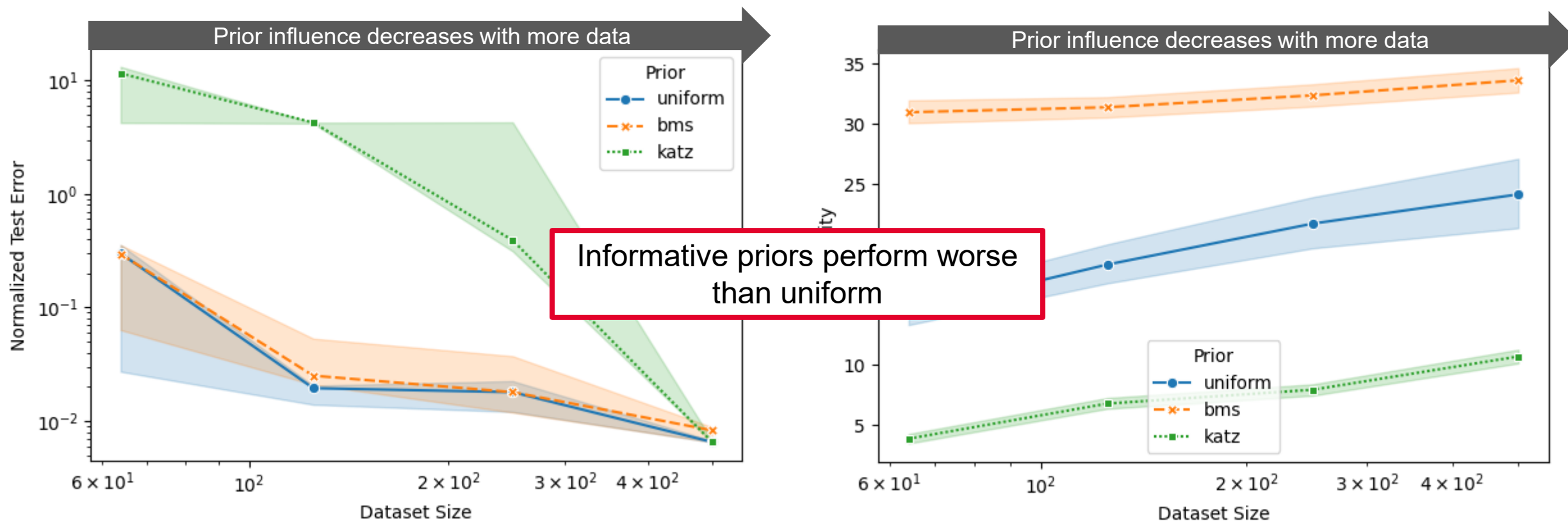
$\log P(M)$ for representative equations (higher = more likely)

	Uniform	BMS	Katz
$\sin(x_0) + \sin(x_1)$	0	-2.80	-18.03
$\sin(\sin(x_0 + x_1))$	0	-2.80	-20.23

We ran into an issue...

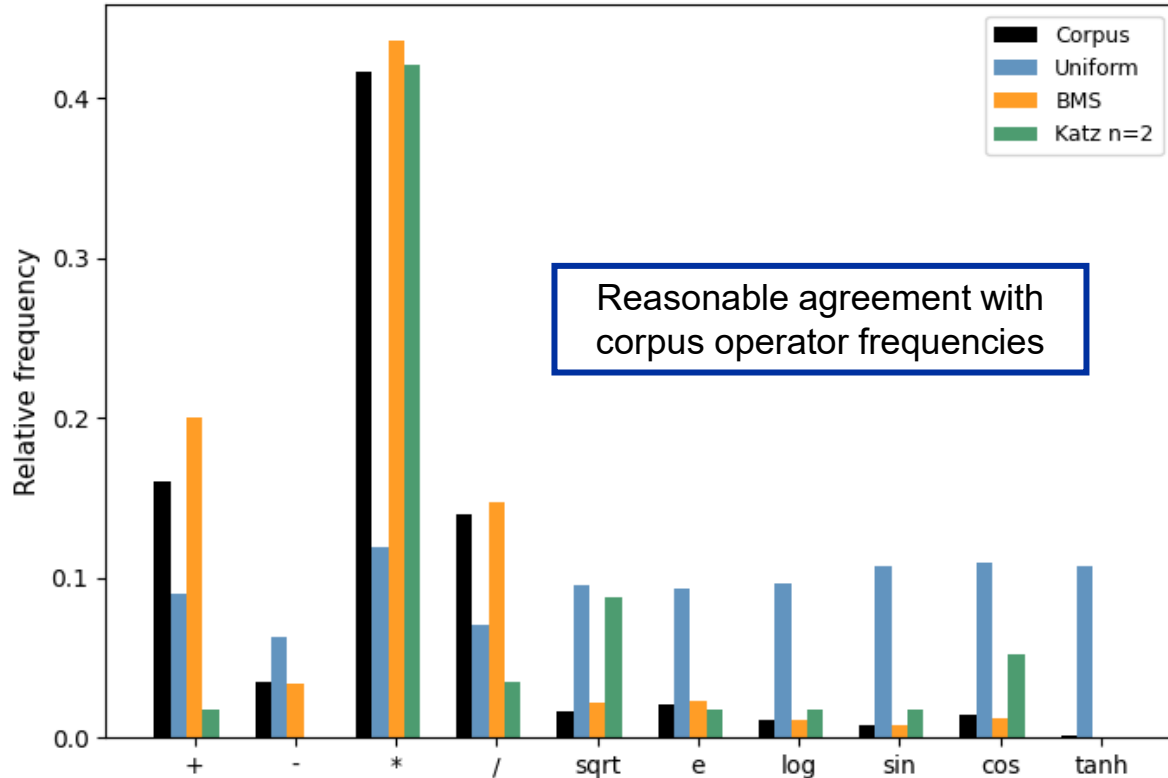
PySIPS applied to 12 Feynman “hard” datasets + 10% noise, 40 repetitions each

- Benchmark equations were not included in the prior corpus

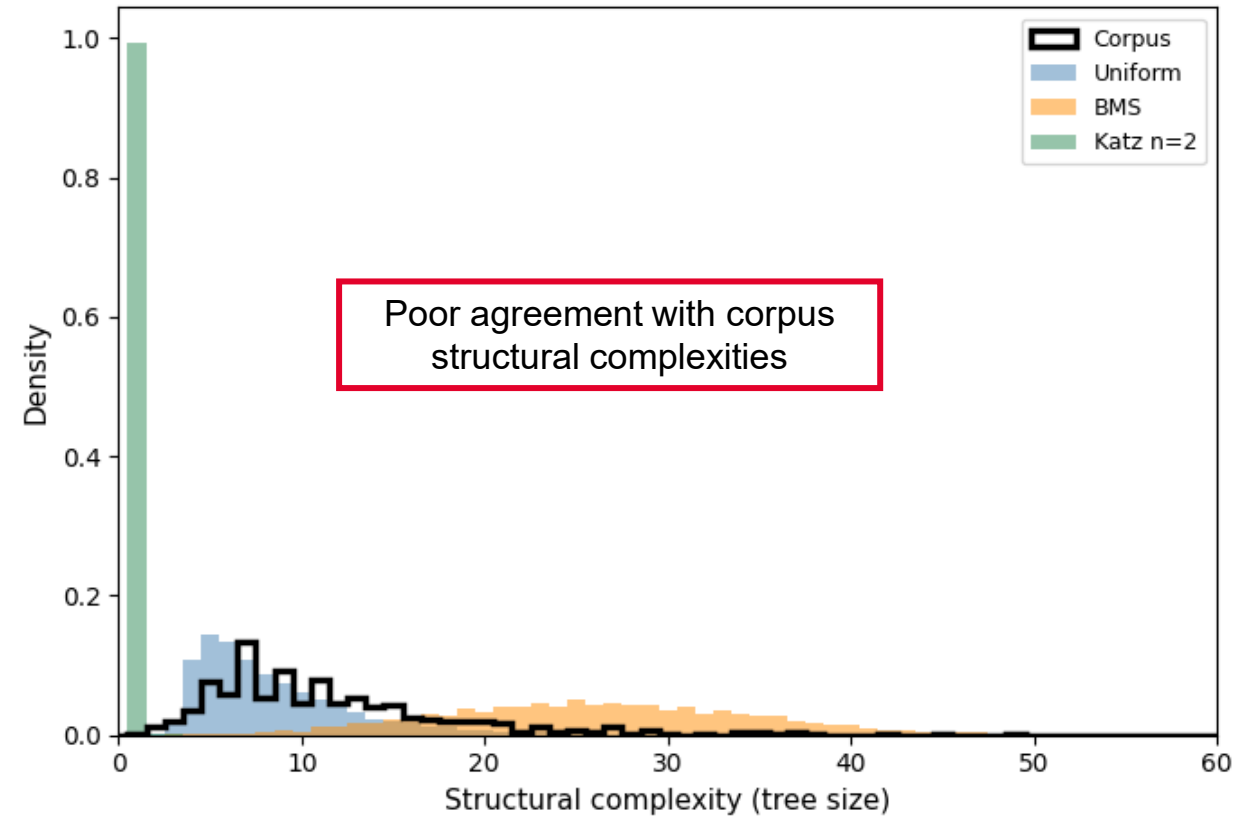


Comparing Priors to Training Corpus

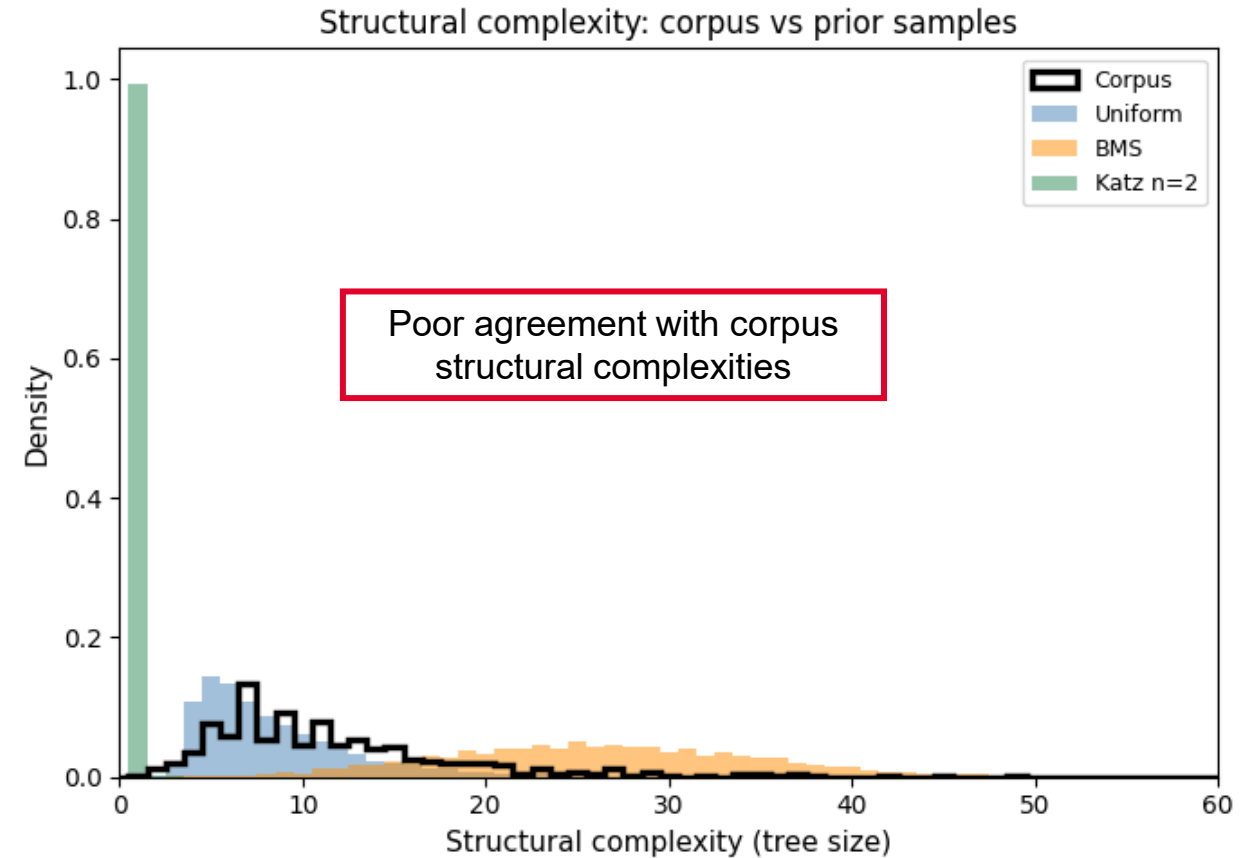
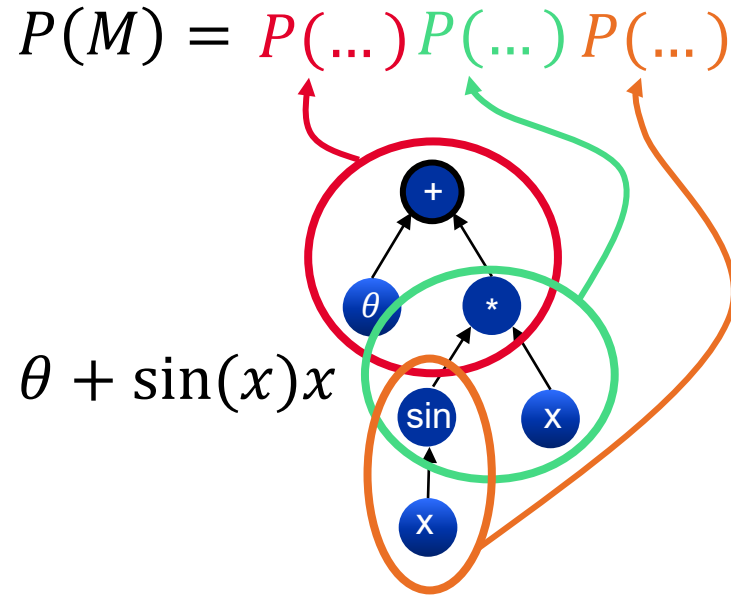
Operator frequencies: corpus vs prior samples



Structural complexity: corpus vs prior samples



Comparing Priors to Training Corpus



- These priors do not explicitly aim to match the corpus complexity distribution

Complexity calibration

- Given a fixed complexity the priors function well
- Inter-complexity comparisons are biased
- **Solution:** complexity calibration
 - Tilt distribution to match the corpus complexity distribution
 - Minimum KL-divergence projection
 - Keeps within-complexity comparisons equal
 - Normalizes each complexity

$\log P(M)$ for representative equations (higher = more likely)

	Uniform	BMS	Katz
$\sin(x_0) + \sin(x_1)$	0	-2.80	-18.03
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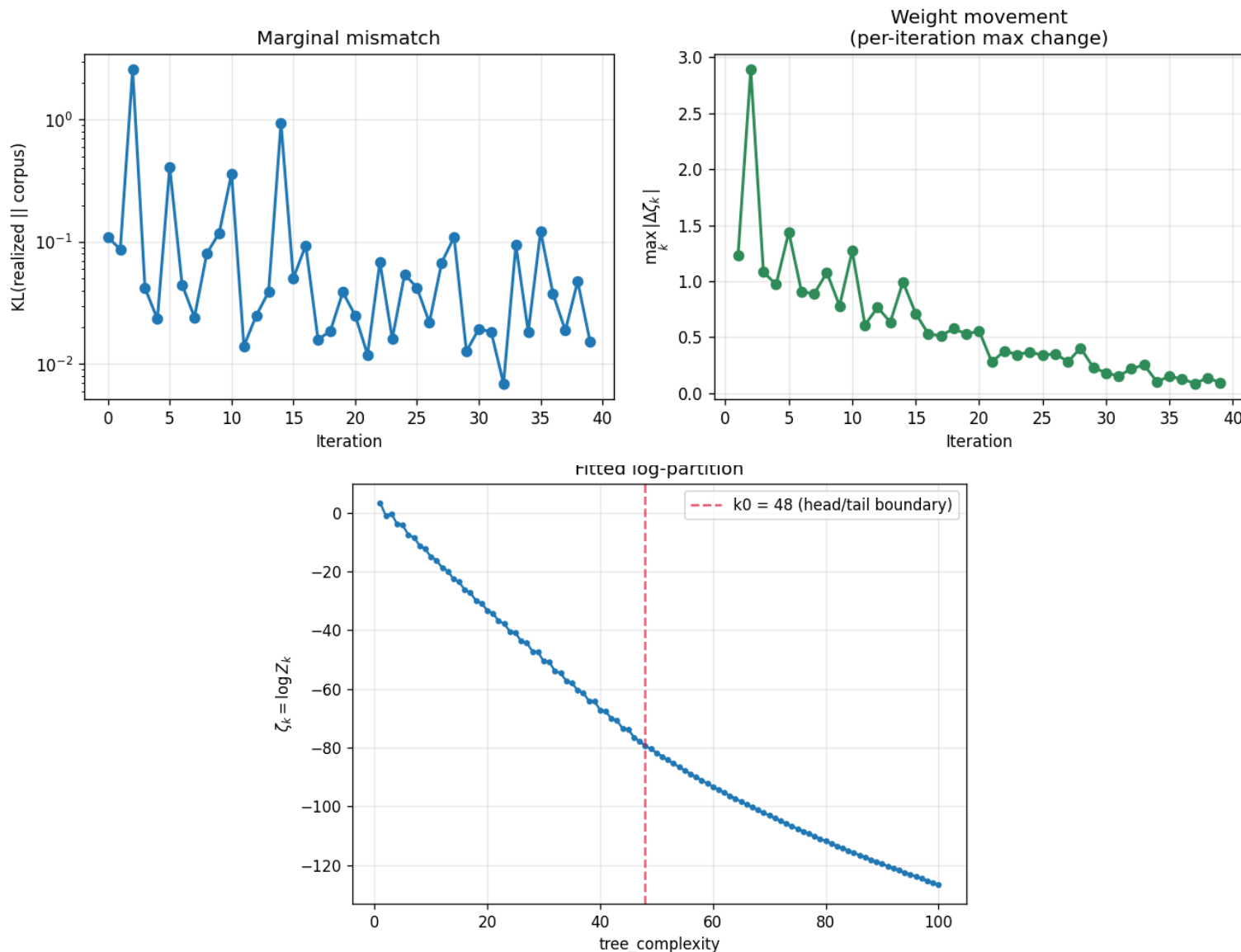
$$Z_k = \sum_{M:|M|=k} P_{base}(M)$$

$$P_{cal}(M) = \frac{p_{corpus}(|M|)P_{base}(M)}{Z_{|M|}}$$

Complexity calibration - Fitting Z_k

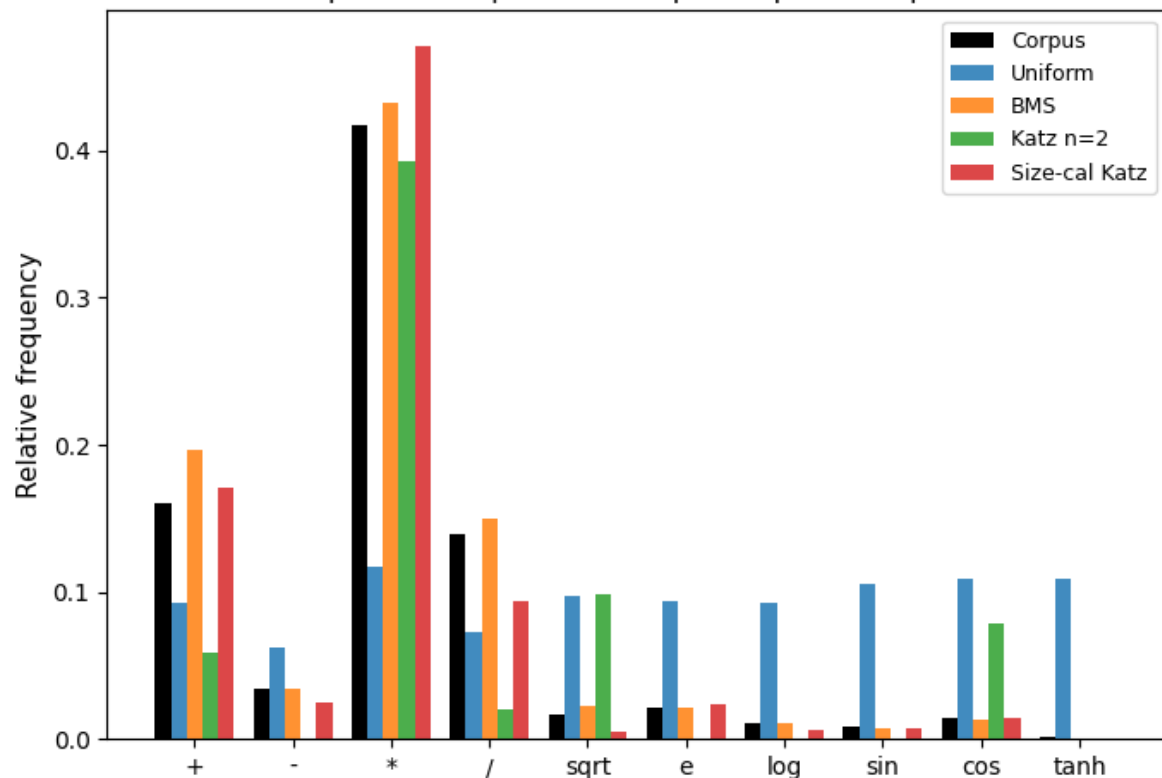
- ✗ Analytical
 - Simplification process is too complex
- ✗ Brute force
 - Too many to trees to enumerate
- ✗ Hybrid
- ✓ Inverse
 - Iterative fit for Z_k based on repeated sampling of tilted distribution
 - Simple, but noisy convergence
 - Linear + Log tail fit

$$Z_k = \sum_{M: |M|=k} P_{base}(M)$$



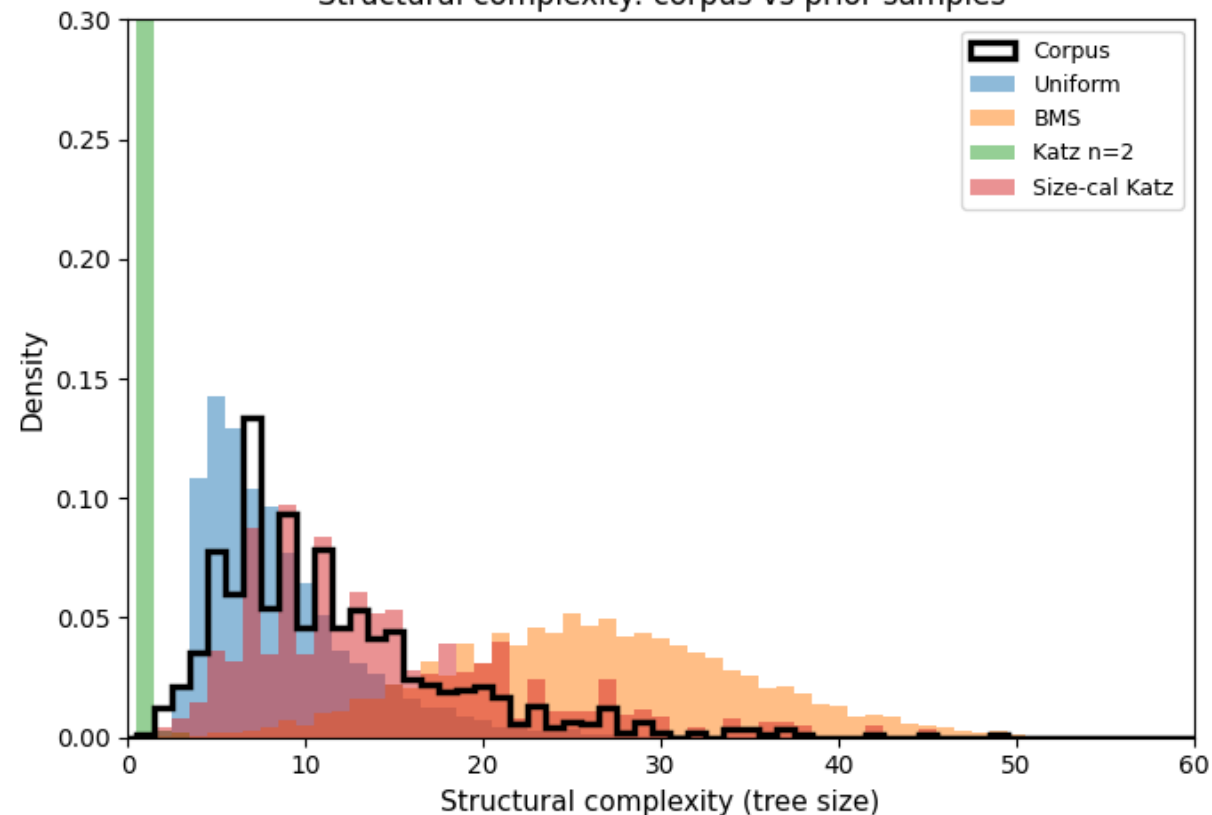
Size-Calibrated Katz Prior

Operator frequencies: corpus vs prior samples



Good agreement with corpus
operator frequencies

Structural complexity: corpus vs prior samples

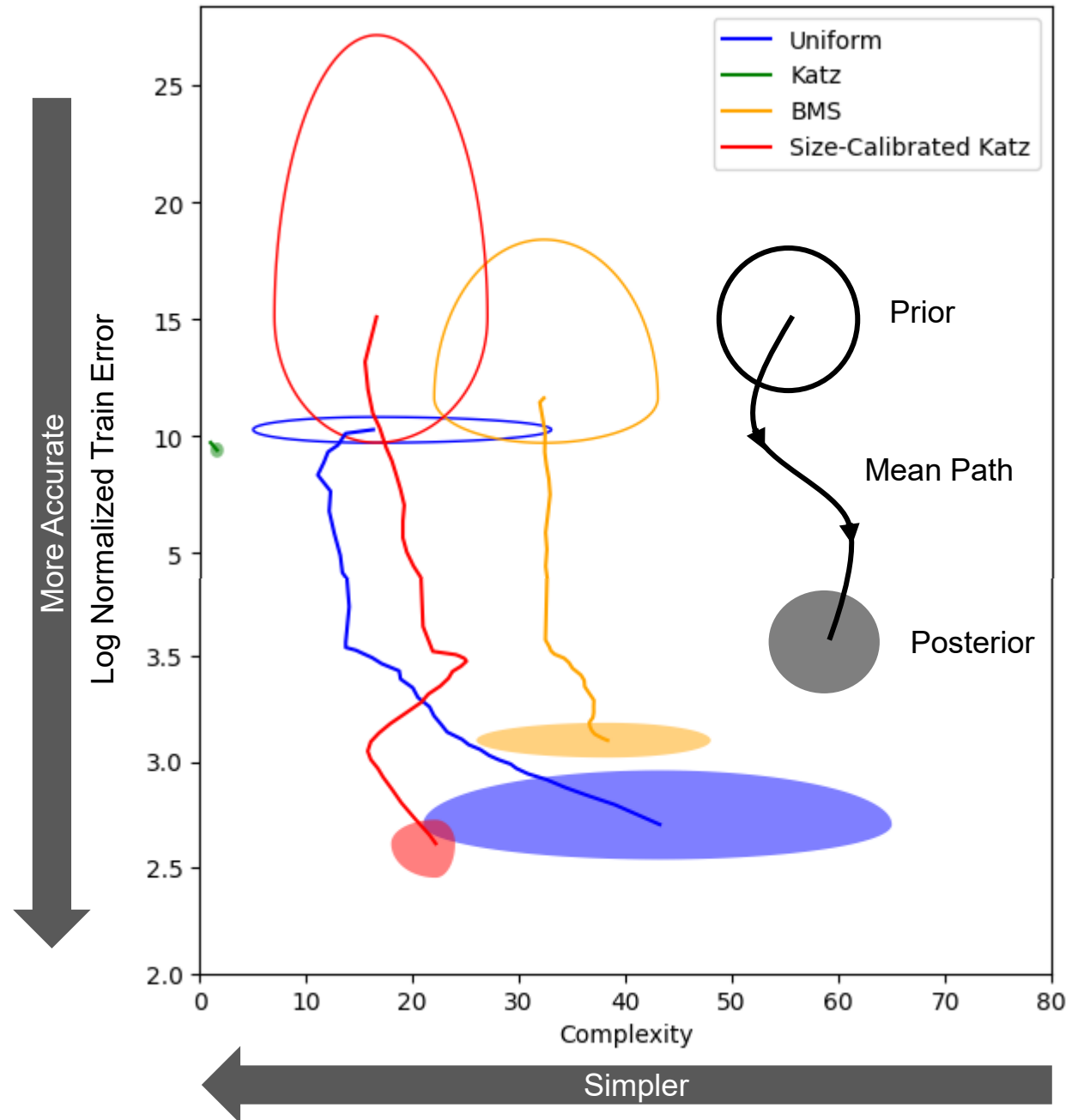


Good agreement with corpus
complexity

Results

SMC in Action

- Results for single run on one of the Feynman benchmark datasets
- Size-Calibrated Katz finds a simpler, better-fitting expressions



Equations Produced

Target Equation: $Q = knT\ln\left(\frac{V_2}{V_1}\right)$

Prior	Posterior Samples		
Size-Cal. Katz	$\theta_0 n T \ln\left(\frac{V_2}{V_1}\right)$	$\theta_0 n T \ln\left(\theta_1 \frac{V_2}{V_1}\right)$	$\theta_0 n T \ln\left(\theta_1 \frac{V_2}{V_1} \frac{(2n + nV_1)}{n}\right)$
Uniform	$\theta_0 - n\left(\theta_1 + \theta_2 T\left(\frac{\ln(V_2)}{\theta_0} + \ln\left(\frac{V_1}{\theta_0}\right)\right)\right)$	$\theta_0 n T \ln\left(\frac{V_2}{V_1}\right)$	$\frac{\theta_0 n T (V_2 - V_1)}{V_2 + V_1 + \theta_1 (V_2 - V_1)^2 + \theta_2 V_2 V_1 \ln(V_1)}$
Katz	V_1	V_1	$\theta_0 n T (V_2 - V_1)$
BMS	$\frac{\theta_0 n}{V_2} + \frac{\theta_1 n T V_1}{V_2 + V_1} - \frac{V_2}{V_1} \left(\frac{\theta_0 n}{V_2} + \frac{\theta_1 n T V_1}{V_2 + V_1}\right)$	$\frac{8nT\ln\left(\frac{V_2}{V_1}\right)}{\theta_0 + 2V_2}$	$\theta_0 T \left(\frac{V_2}{\theta_1 V_1} - \theta_1 V_1 + n * \theta_1 V_1 - \ln\left(-\frac{V_2}{\theta_1 V_1} + \theta_1 V_1\right)\right)$

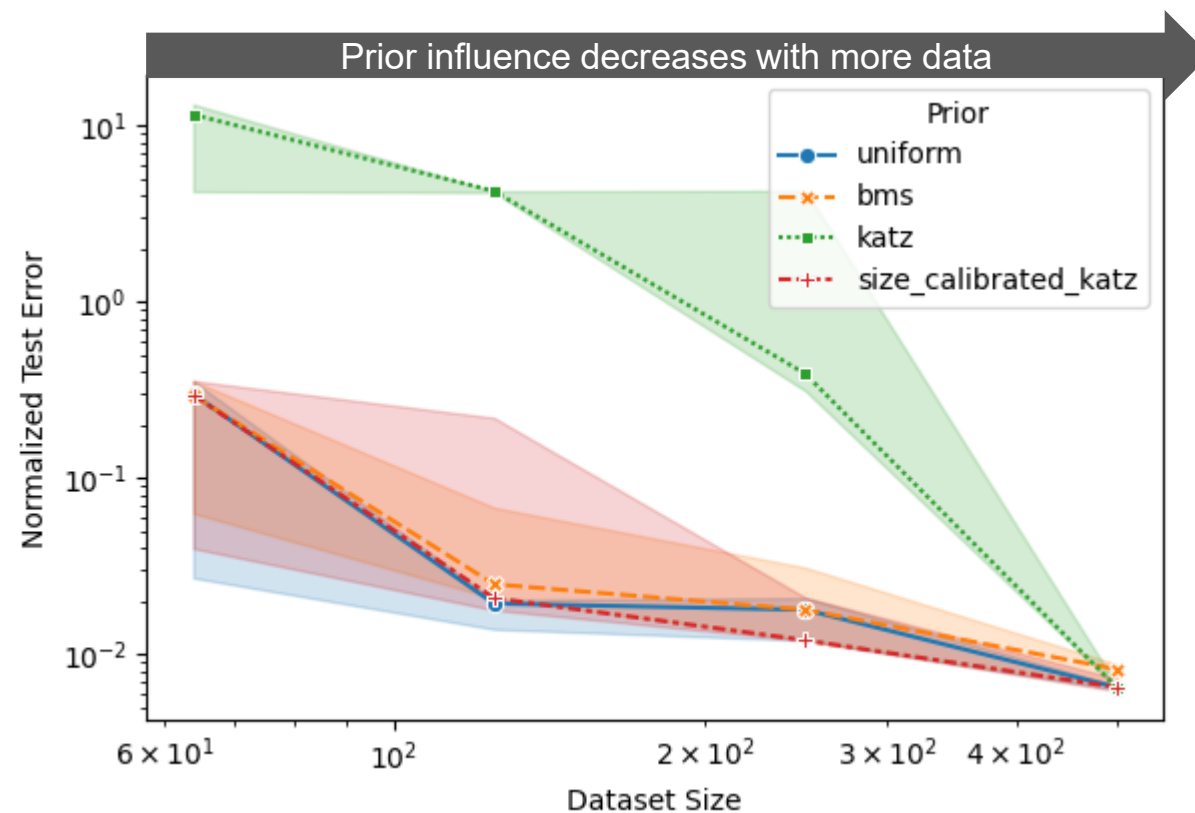
Benchmark Results

Size-calibrated Katz

Generalization Ability

- Performs similar to uniform and BMS
- Improves over Katz

Generalization Ability (Test Error)



Benchmark Results

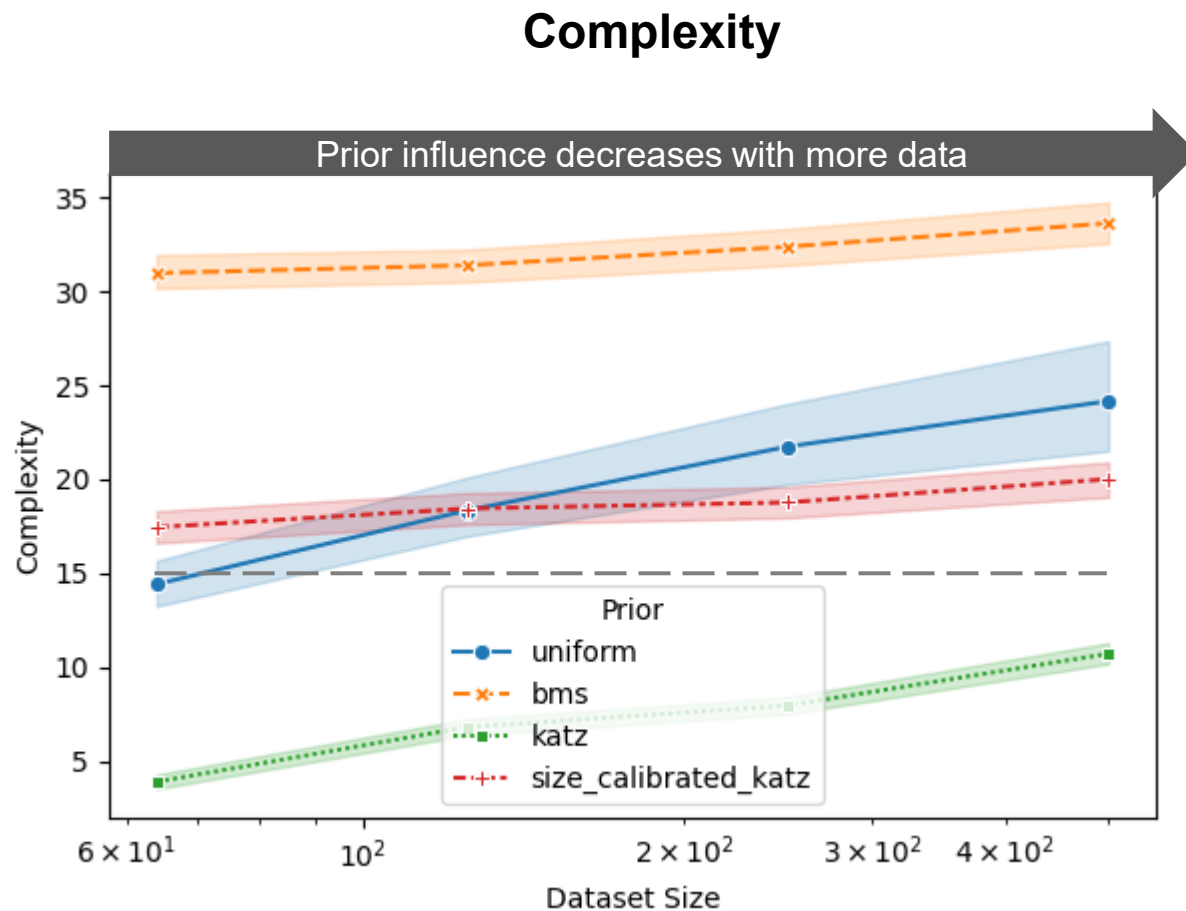
Size-calibrated Katz

Generalization Ability

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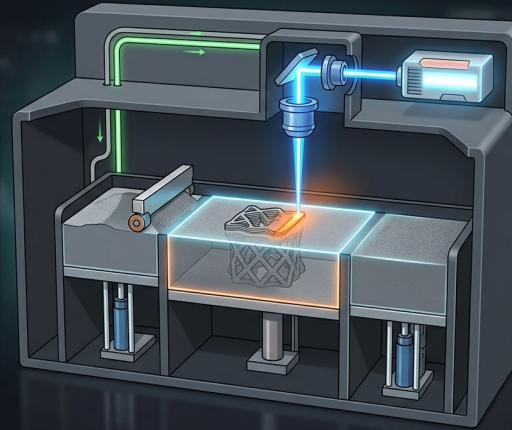
Complexity

- Produces similar sized equations across dataset sizes

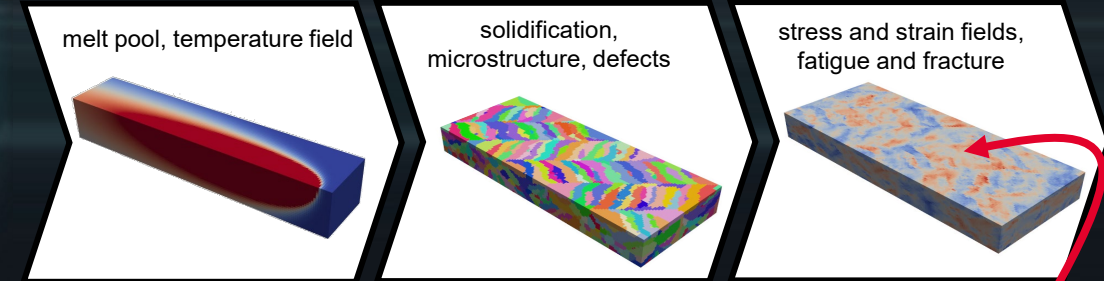


Future Work: Additive Manufacturing Application

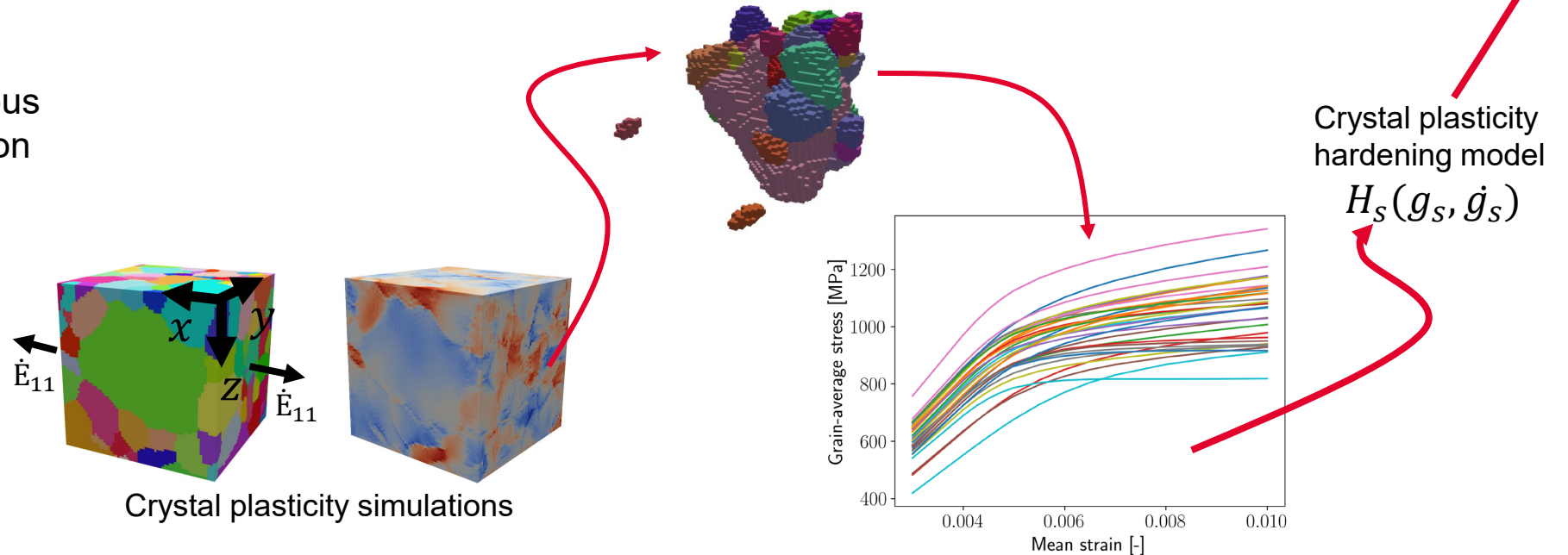
- Qualification and certification of powder bed fusion
- Defining hardening laws for crystal plasticity models



MODELING PIPELINE



- Prior:
 - Bootstrapped with Feynman/Wikipedia corpus
 - Fine tune with literature on hardening models
- Interpretability judged by subject mater experts



Thank you!

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PYSIPS
Symbolic Inference via
Posterior Sampling

